

BRAF fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes BRAF gene fusions from the msk_impact_50k_2026 study, covering 179 fusion events. 151/179 fusions (84.4%) were in-frame. 163/179 fusions (91.1%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0133676$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 84.4% in-frame and 91.1% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 179 analyzed fusion events, identifying 67 distinct fusion partner genes. The most frequent partner was KIAA1549, observed in 43 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions retained the domain, $p=0.0133676$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.04995.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 126/151 (83.4%) of in-frame fusions disrupted the domain, $p=0.318252$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.656934.

Cutpoint detection

Cutpoint detection scanned 178 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 439 aa (permutation-corrected $p=0.000999001$, statistically significant at $\alpha=0.05$). This is 19 aa from the nearest configured domain boundary at 458 aa.

Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: Algorithm failed: ValueError: params['group_field'] is required

Composite evidence score

Composite evidence score reported: confidence_certainty sub-score excluded: no applicable confidence_stats MLE/CI result was supplied.

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: Algorithm failed: ValueError: joint_partner requires a GeneConfig with gene_pair

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIAA1549	43	0.24022346368715083	0.13014640731432997	0.018247805771708193	0.8884995221408092	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2683645893446564	1
SND1	17	0.09497206703910614	0.13014640731432997	0.018247805771708193	0.8642224012892828	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.21590127032005374	2
SCRIB	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	1.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2087357118183949	3
SVOPL	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	1.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2087357118183949	4
TRIM33	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	1.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2087357118183949	5
EPB41	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.997716894977169	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20835519431458974	6
PJA2	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.997716894977169	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20835519431458974	7
RBM33	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.997716894977169	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20835519431458974	8
TRIM24	8	0.0446927374301676	0.13014640731432997	0.018247805771708193	0.886986301369863	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20293547713050428	9
MKRN1	10	0.055865921787709494	0.13014640731432997	0.018247805771708193	0.8447488584474886	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19962029809595586	10
CDK5RAP2	4	0.0223463687150838	0.13014640731432997	0.018247805771708193	0.906392694063927	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19872108634115368	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ATF7	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	12
DOCK4	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	13
GIPC2	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	14
GTF2I	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	15
KCND2	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	16
NFIA	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	17
OSBPL9	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	18
PRKAR2B	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	19
RRBP1	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	20
SETBP1	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	21
SLC45A3	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	22
SPOCK1	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	23
ZC3HAV1	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ZNF207	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8949771689497716	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19123190664335682	25
NRF1	2	0.0111731843575419	0.13014640731432997	0.018247805771708193	0.8812785388127854	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19081099901344942	26
EXOC4	2	0.0111731843575419	0.13014640731432997	0.018247805771708193	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18852789399061837	27
AKAP9	2	0.0111731843575419	0.13014640731432997	0.018247805771708193	0.8664383561643836	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18833763523871577	28
MAD1L1	2	0.0111731843575419	0.13014640731432997	0.018247805771708193	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18814737648681318	29
CCDC6	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18666569659769475	30
CMAS	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18666569659769475	31
ERC1	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18666569659769475	32
MINDY4	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18666569659769475	33
PHTF2	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18666569659769475	34
SNX8	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.8675799086757991	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18666569659769475	35
ABCC1	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18628517909388956	36
CAPZA2	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18628517909388956	37

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
OSBPL7	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18628517909388956	38
PRIM2	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18628517909388956	39
PRKAR1B	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18628517909388956	40
PTPRZ1	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18628517909388956	41
RGS3	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.865296803652968	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18628517909388956	42
CLIP1	2	0.0111731843575419	0.13014640731432997	0.018247805771708193	0.8059360730593608	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1782539213878768	43
TMPRSS2	4	0.0223463687150838	0.13014640731432997	0.018247805771708193	0.7678843226788432	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17563635777697303	44
ABCC2	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1661177513922153	45
ITGB5	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1661177513922153	46
MKLN1	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1661177513922153	47
NEO1	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1661177513922153	48
UBE2H	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1661177513922153	49
WEE2	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.7442922374429224	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1661177513922153	50

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
AGK	14	0.0782122905027933	0.13014640731432997	0.018247805771708193	0.5808871493803001	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16309213615645238	51
LUC7L2	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.6529680365296804	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1508970512400083	52
AGAP3	6	0.0335195530726257	0.13014640731432997	0.018247805771708193	0.4330289193302892	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12355151867139468	53
GTF2IRD1	3	0.01675977653631285	0.13014640731432997	0.018247805771708193	0.4657534246575342	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12341901071383124	54
FAM131B	3	0.01675977653631285	0.13014640731432997	0.018247805771708193	0.2891933028919331	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.09399232375289772	55
CARMIL1	2	0.0111731843575419	0.13014640731432997	0.018247805771708193	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08578816796322114	56
CLEC2L	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08392597057029749	57
DBI	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08392597057029749	58
PLPP1	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08392597057029749	59
RBMS3	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08392597057029749	60
TBXAS1	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.25114155251141557	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08392597057029749	61
CUL1	5	0.0279329608938547	0.13014640731432997	0.018247805771708193	0.17305936073059358	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.0783610615118551	62
KLHL12	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04206904515172824	63

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
LRBA	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04206904515172824	64
PAK2	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04206904515172824	65
PWWP2A	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04206904515172824	66
TRA2B	1	0.00558659217877095	0.13014640731432997	0.018247805771708193	0.0	None	['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04206904515172824	67

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1	17	146	0	15	None	0.36712402624074814	0.43518719238968984
238	18	145	0	15	None	0.3697885050600375	0.43204659308648946
287	18	145	1	14	1.7379310344827585	1.0	-0.0
303	19	144	1	14	1.8472222222222223	1.0	-0.0
327	32	131	5	10	0.48854961832061067	0.3143387647670962	0.5026020577787302
380	72	91	6	9	1.1868131868131868	0.7933319684125757	0.10054504486238983
381	115	48	7	8	2.738095238095238	0.07931888947474959	1.1006233748798022
393	138	25	7	8	6.308571428571429	0.001556599022521673	2.807823246605722
400	139	24	7	8	6.619047619047619	0.0012299959408670453	2.9100963217817015
438	152	11	8	7	12.090909090909092	0.0001283182416528584	3.891711600164127
439	158	5	9	6	21.066666666666666	4.522271339496731e-05	4.344643383231083
744	158	5	10	5	15.8	0.000419446170360627	3.3773237664571667

domain_disruption tables

frame_domain_contingency_table

126	21
25	6

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ras_binding	RAS-binding domain	178	31	0	147	0.0
cysteine_rich	Cysteine-rich domain	178	31	1	146	0.006359065365276546

domain_retention tables

frame_domain_contingency_table

142	21
9	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	178	163	0	15	0.0

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ABCC1	1
ABCC2	1
AGAP3	6
AGK	14
AKAP9	2
ATF7	1
CAPZA2	1
CARMIL1	2
CCDC6	1
CDK5RAP2	4
CLEC2L	1
CLIP1	2
CMAS	1
CUL1	5
DBI	1

Partner_gene	Event_count
DOCK4	1
EPB41	1
ERC1	1
EXOC4	2
FAM131B	3
GIPC2	1
GTF2I	1
GTF2IRD1	3
ITGB5	1
KCND2	1
KIAA1549	43
KLHL12	1
LRBA	1
LUC7L2	1
MAD1L1	2
MINDY4	1
MKLN1	1
MKRN1	10
NEO1	1
NFIA	1
NRF1	2
OSBPL7	1
OSBPL9	1
PAK2	1
PHTF2	1
PJA2	1
PLPP1	1
PRIM2	1
PRKAR1B	1
PRKAR2B	1
PTPRZ1	1
PWWP2A	1
RBM33	1
RBMS3	1
RGS3	1

Partner_gene	Event_count
RRBP1	1
SCRIB	1
SETBP1	1
SLC45A3	1
SND1	17
SNX8	1
SPOCK1	1
SVOPL	1
TBXAS1	1
TMPRSS2	4
TRA2B	1
TRIM24	8
TRIM33	1
UBE2H	1
WEE2	1
ZC3HAV1	1
ZNF207	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021458-T01-IM6-1	P-0021458-T01-IM6		ABCC1-BRAF	ABCC1	in-frame	three_prime	140493152	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC1 (NM_004996) - BRAF (NM_004333) rearrangement: t(7;16)(q34,p13.11)(chr7:g.140493152::chr16:g.16116105) Protein Fusion: in frame {ABCC1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 4621-T01-I M6-2	P-0034 621-T01-IM6		ABCC2-BRAF	ABCC2	unknown	three_prime	140495500	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC2 (NM_000392) - BRAF (NM_004333) rearrangement: t(7;10)(q34;24.2)(chr7:g.140495500::chr10:g.101559038) Protein Fusion: mid-exon {ABCC2:BRAF}	NA
EVT-P-005 2582-T01-I M6-3	P-0052 582-T01-IM6		AGAP3-BRAF	AGAP3	unknown	three_prime	150835371	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1637:AGAP3_c.1177+1690:BRAFinv Protein Fusion: mid-exon {AGAP3:BRAF}	NA
EVT-P-005 0574-T02-I M6-4	P-0050 574-T02-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	150823608	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1222-2059:AGAP3_c.1140+900:BRAFinv Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-003 9799-T01-I M6-5	P-0039 799-T01-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	150821389	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1221+416:AGAP3_c.1314+504:BRAFinv Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-005 0931-T01-I M6-6	P-0050 931-T01-IM6		AGK-BRAF	AGK	in-frame	three_prime	141278652	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-14294:AGK_c.980+911:BR AFinv Protein Fusion: in frame {AGK:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 2835-T03-I M6-7	P-0042 835-T0 3-IM6		AGK- BRAF	AGK	in-frame	three_prime	141279 302	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cytochrome-rich domain			AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-004 2835-T02-I M6-8	P-0042 835-T0 2-IM6		AGK- BRAF	AGK	in-frame	three_prime	141279 302	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cytochrome-rich domain			AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-005 0635-T01-I M6-9	P-0050 635-T0 1-IM6		AKAP 9-BRAF	AKAP9	in-frame	three_prime	140487 501	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.6613-1489:AKAP9_c.1141-117:BRAFInv Protein Fusion: in frame {AKAP9:BRAF}	NA
EVT-P-003 0237-T03-I M6-10	P-0030 237-T0 3-IM6		AKAP 9-BRAF	AKAP9	in-frame	three_prime	140492 357	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.3318+1378:AKAP9_c.1140+1751:BRAFInv Protein Fusion: in frame {AKAP9:BRAF}	NA
EVT-P-002 3821-T01-I M6-12	P-0023 821-T0 1-IM6		ATF7- BRAF	ATF7	in-frame	three_prime	140483 233	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ATF7 (NM_006856) - BRAF (NM_004333) Rearrangement : t(7;12)(q34;q13.13)(chr7:g.140483233::chr12:g. 53913299) Protein Fusion: in frame {ATF7:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0007861-T01-IM5-13	P-0007861-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140490697	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1326+220:AGAP3_c.1141-3313:BRAF Protein fusion: in frame (AGAP3-BRAF)	NA
EVT-P-0021519-T02-IM6-14	P-0021519-T02-IM6		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493112	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) Rearrangement : c.1326+142:AGAP3_c.1140+996:BRAF Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-0009757-T01-IM5-15	P-0009757-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493871	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1222-2154:AGAP3_c.1140+237:BRAF Protein fusion: in frame (AGAP3-BRAF)	NA
EVT-P-0036886-T01-IM6-16	P-0036886-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494309	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+10859:AGK_c.981-42:BRAF Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0025561-T03-IM6-17	P-0025561-T03-IM6		BRAF-AGK	AGK	in-frame	three_prime	140498719	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 8-18): c.980+1288:BRAF_c.101+772:AGK Protein Fusion: in frame {AGK:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0025561-T01-IM6-18	P-0025561-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498874	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) Rearrangement: c.980+1288:BRAF_c.101+772:AGKinv Protein Fusion: in frame {BRAF:AGK}	NA
EVT-P-0024227-T01-IM6-19	P-0024227-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140495070	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+16796:AGK_c.981-803:BRAFinv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0004477-T01-IM5-20	P-0004477-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140495405	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10076:AGK_c.981-1138:BRAFinv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0023378-T01-IM6-21	P-0023378-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498362	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) rearrangement: c.980+1800:BRAF_c.101+14574:AGKinv Protein Fusion: in frame {BRAF:AGK}	NA
EVT-P-0001832-T02-IM6-22	P-0001832-T02-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494489	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exon 1 fused in-frame to BRAF exons 8-18): c.102-572:AGK_c.981-222:BRAFinv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0000913-T02-IM5-23	P-0000913-T02-IM5		BRAF-AGK	AGK	in-frame	three_prime	140496386	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BRAFinv Protein fusion: in frame (AGK-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 9531-T01-I M6-24	P-0029 531-T01-IM6		BRAF-AGK	AGK	unknown	three_prime	140500233	7	303	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 7-18): c.101+4572:AGK_c.909:BRAFinv Protein Fusion: mid-exon {AGK:BRAF}	NA
EVT-P-003 0746-T01-I M6-25	P-0030 746-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140497533	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BRAFinv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-001 0945-T01-I M5-26	P-0010 945-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140497610	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFinv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-004 9035-T01-I M6-85	P-0049 035-T01-IM6		BRAF-CLEC2L	CLEC2L	in-frame	five_prime	139215088	18	767	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			BRAF (NM_004333) - CLEC2L (NM_001080511) rearrangement: c.1177+1549:BRAF_c.191-5932:CLEC2Linv Protein Fusion: in frame {BRAF:CLEC2L}	NA
EVT-P-000 4795-T01-I M5-87	P-0004 795-T01-IM5		BRAF-CUL1	CUL1	in-frame	three_prime	140493811	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1(NM_003592) - BRAF (NM_004333) Rearrangement :c.790-427: CUL1_c.1140+297:BRAFinv Protein fusion: in frame (CUL1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 2257-T01-I M6-88	P-0022 257-T01-IM6		BRAF-EPB41	EPB41	in-frame	three_prime	140482687	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EPB41 (NM_001166005) - BRAF (NM_004333) fusion (EPB41 exons 1-5 fused in-frame with BRAF exons 11-18): t(1;7)(p35.4;q34)(chr1:g.29424227::chr7:g.140482687) Protein Fusion: in frame {EPB41:BRAF}	NA
EVT-P-000 1988-T01-I M3-89	P-0001 988-T01-IM3		BRAF-FAM131B	FAM131B	in-frame	three_prime	140489629	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) Deletion : c.175-103:FAM131B_c.1141-2245:BRAFdel Protein fusion: in frame (FAM131B-BRAF)	NA
EVT-P-001 0494-T01-I M5-90	P-0010 494-T01-IM5		BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655) -BRAF (NM_004333) Rearrangement : t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) Protein fusion: in frame (GIPC2-BRAF)	NA
EVT-P-005 4046-T01-I M6-93	P-0054 046-T01-IM6		BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	73984431	18	767	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-1749:BRAF_c.2320+10446:GTF2IRD1inv Protein Fusion: in frame {BRAF:GTF2IRD1}	NA
EVT-P-000 0828-T03-I M6-94	P-0000 828-T03-IM6		BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	73988166	18	767	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-597:BRAF_c.2320+14181:GTF2IRD1inv Protein Fusion: in frame {BRAF:GTF2IRD1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 5847-T02-I M6-95	P-0015 847-T0 2-IM6		BRAF- ITGB5	ITGB5	out-of-frame	five_prime	140497019	8	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - ITGB5 (NM_002213) Rearrangement : t(3,7)(q21.2;q34)(chr3:g.124579522::chr7:g.140497019) Protein Fusion: out of frame {BRAF:ITGB5}	NA
EVT-P-001 4377-T01-I M6-99	P-0014 377-T0 1-IM6		BRAF- NFIA	NFIA	in-frame	three_prime	140486673	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NFIA (NM_001145512) - BRAF (NM_004333) rearrangement: t(1;7)(p31.3;q34)(chr1:g.61836194::chr7:g.140486673) Protein fusion: in frame (NFIA-BRAF)	NA
EVT-P-001 0398-T01-I M5-100	P-0010 398-T0 1-IM5		BRAF- OSBP L9	OSBP L9	in-frame	three_prime	140484912	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL9 (NM_148909) - BRAF (NM_004333) Rearrangement : t(1;7)(p32.3;q34)(chr1:g.52228623::chr7:g.140484912) Protein fusion: in frame (OSBPL9-BRAF)	NA
EVT-P-000 3502-T01-I M5-101	P-0003 502-T0 1-IM5		BRAF- PJA2	PJA2	in-frame	five_prime	140482379	10	438	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA Protein fusion: in frame (BRAF-PJA2)	NA
EVT-P-003 6870-T01-I M6-102	P-0036 870-T0 1-IM6		BRAF- PRIM2	PRIM2	in-frame	three_prime	140493232	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRIM2 (NM_000947) - BRAF (NM_004333) Rearrangement : t(6;7)(p11.1;q34)(chr6:g.57410815::chr7:g.140493232) Protein Fusion: in frame {PRIM2:BRAF}	NA
EVT-P-000 9246-T01-I M5-103	P-0009 246-T0 1-IM5		BRAF- RBM33	RBM33	in-frame	three_prime	140482661	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBM33 (NM_053043) - BRAF (NM_004333) rearrangement: c.171+526:RBM33_c.1314+160:BRAFInv Protein fusion: in frame (RBM33-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 6313-T01-I M6-104	P-0016 313-T0 1-IM6		BRAF- SLC4 5A3	SLC 45A 3	unknown	three_prime	140486 608	9	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SLC45A3 (NM_033102) - BRAF (NM_004333) fusion (SLC45A3 exons 1 fused to BRAF exons 10-18): t(1;7)(q 32.1;q34)(chr1:g.20563845 2::chr7:g.140486608) Transcript Fusion {SLC45A3:BRAF}	NA
EVT-P-006 4764-T01-I M7-105	P-0064 764-T0 1-IM7		BRAF- SND1	SND 1	in-frame	five_prime	127357 391	18	767	True	retained	1.0	False	Protein kinase domain; RAS- binding domain; Cysteine- rich domain			BRAF (NM_004333) - SND1 (NM_014390) fusion: c.1140 +819:BRAF_c.1039-3950:S ND1inv Protein Fusion: in frame {BRAF:SND1}	NA
EVT-P-002 9981-T01-I M6-106	P-0029 981-T0 1-IM6		BRAF- SPOCK1	SPO CK1	in-frame	three_prime	140483 862	10	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SPOCK1 (NM_004598) - BRAF (NM_004333) fusion: t(5;7)(q31.2;q34)(chr5:g.136 401840::chr7:g.140483862) Protein Fusion: in frame {SPOCK1:BRAF}	NA
EVT-P-006 4223-T01-I M7-107	P-0064 223-T0 1-IM7		BRAF- TBXAS1	TBX AS1	out-of-frame	five_prime	139614 105	18	767	True	retained	1.0	False	Protein kinase domain; RAS- binding domain; Cysteine- rich domain			BRAF (NM_004333) - TBXAS1 (NM_001061) rearrangement: c.981-245:B RAF_c.336+2985:TBXAS1i nv Protein Fusion: out of frame {BRAF:TBXAS1}	NA
EVT-P-002 1469-T01-I M6-108	P-0021 469-T0 1-IM6		BRAF- TRIM3	TRI M33	in-frame	three_prime	140481 644	11	439	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		TRIM33 (NM_015906) - BRAF(NM_004333) rearrangement: t(1;7)(p13.2; q34)(chr1:g.114963985::chr 7:g.140481644) Protein Fusion: in frame {TRIM33:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 2835-T03-I M6-109	P-0042 835-T03-IM6		BRAF-WEE2	WEE2	unknown	five_prime	140498912	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - WEE2 (NM_001105558) rearrangement: c.980+1250 :BRAF_c.1305:WEE2inv Protein Fusion: mid-exon {BRAF:WEE2}	NA
EVT-P-001 5985-T03-I M6-111	P-0015 985-T03-IM6		CAPZA2-BRAF	CAPZA2	in-frame	three_prime	140491425	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CAPZA2 (NM_006136) - BRAF (NM_004333) fusion: c.40-938:CAPZA2_c.1140+2683:BRAFinv Protein Fusion: in frame {CAPZA2:BRAF}	NA
EVT-P-003 3764-T04-I M6-112	P-0033 764-T04-IM6		CARMIL1-BRAF	CARMIL1	in-frame	three_prime	25554753	18	767	True	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion: t(6;7)(p22.2;q34)(chr6:g.25554753::chr7:g.140491509) Protein Fusion: in frame {LRRC16A:BRAF}	NA
EVT-P-003 0796-T01-I M6-113	P-0030 796-T01-IM6		CARMIL1-BRAF	CARMIL1	in-frame	three_prime	25580104	18	767	True	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion (LRRC16A exons 1-29 fused to BRAF exons 9-18): t(6;7)(p22.2;q34)(chr6:g.25580104::chr7:g.140491645) Protein Fusion: in frame {LRRC16A:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 5834-T01-I M5-114	P-0005 834-T0 1-IM5		CCDC6-BRAF	CCDC6	in-frame	three_prime	140488401	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CCDC6 (NM_005436) -BRAF (NM_004333) Rearrangement : t(10;7)(chr10q21.2;q34)(chr10:g.61627617::chr7:g.140488401) Protein fusion: in frame (CCDC6-BRAF)	NA
EVT-P-003 8194-T01-I M6-115	P-0038 194-T0 1-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140484954	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) -BRAF (NM_004333) rearrangement: t(7;9)(q34;q33.2)(chr7:g.140484954::chr9:g.123187156) Protein Fusion: in frame {CDK5RAP2:BRAF}	NA
EVT-P-000 4203-T01-I M5-116	P-0004 203-T0 1-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140491585	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) -BRAF (NM_004333) rearrangement: t(7;9)(7q34;q33.2)(chr7:140491585::chr9:123251876) Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-000 5461-T02-I M5-117	P-0005 461-T0 2-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140481684	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) -BRAF (NM_004333) rearrangement : t(7;9)(7q34;q33.2)(chr7:140481684::chr9:123255452) Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-001 4969-T01-I M6-118	P-0014 969-T0 1-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	five_prime	140493700	8	380	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) -CDK5RAP2 (NM_018249) Rearrangement : t(7;9)(q34;q33.3)(chr7:g.140493700::chr9:g.123269531) Protein fusion: in frame (BRAF-CDK5RAP2)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 0461-T01-I M6-119	P-0040 461-T0 1-IM6		CLIP1-BRAF	CLIP1	in-frame	three_prime	140488218	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140488218::chr12:g.122833264) Protein Fusion: in frame {CLIP1:BRAF}	NA
EVT-P-000 7266-T02-I M6-120	P-0007 266-T0 2-IM6		CLIP1-BRAF	CLIP1	in-frame	three_prime	140495673	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140495673::chr12:g.122767026) Protein Fusion: in frame {CLIP1:BRAF}	NA
EVT-P-002 4995-T01-I M6-121	P-0024 995-T0 1-IM6		CMAS-BRAF	CMAS	unknown	three_prime	140488046	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) rearrangement: t(7;12)(q34;p12.1)(chr7:g.140488046::chr12:g.22218755) Transcript Fusion {CMAS:BRAF}	NA
EVT-P-003 9648-T01-I M6-122	P-0039 648-T0 1-IM6		CUL1-BRAF	CUL1	in-frame	three_prime	148460538	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215: BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-003 9648-T03-I M6-123	P-0039 648-T0 3-IM6		CUL1-BRAF	CUL1	in-frame	three_prime	148460538	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215: BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006 4088-T01-I M7-124	P-0064 088-T01-IM7		CUL1-BRAF	CUL1	in-frame	threonine	148459544	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+1956:CUL1_c.1141-2067: BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-006 8173-T01-I M7-125	P-0068 173-T01-IM7		CUL1-BRAF	CUL1	in-frame	threonine	148457779	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+191:CUL1_c.1140+2118: BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-005 0196-T01-I M6-126	P-0050 196-T01-IM6		DBI-BRAF	DBI	in-frame	threonine	120128894	18	767	True	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		DBI (NM_001178017) - BRAF (NM_004333) fusion: t(2;7)(q14.2;q34)(chr2:g.120128894::chr7:g.140486284) Protein Fusion: in frame {DBI:BRAF}	NA
EVT-P-006 4588-T01-I M7-127	P-0064 588-T01-IM7		DOCK4-BRAF	DOCK4	out-of-frame	threonine	140483872	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DOCK4 (NM_014705) - BRAF (NM_004333) fusion: c.4650+298:DOCK4_c.1178-915:BRAFdup Protein Fusion: out of frame {DOCK4:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 4833-T02-IM6-128	P-0014 833-T02-IM6		ERC1-BRAF	ERC1	in-frame	three_prime	140490668	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ERC1 (NM_178040) - BRAF (NM_004333) fusion: t(7;12)(q34;p13.33)(chr7:g.140490668::chr12:g.1246645) Protein Fusion: in frame {ERC1:BRAF}	NA
EVT-P-003 6937-T01-IM6-129	P-0036 937-T01-IM6		EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140490241	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) fusion: c.1514+93469:EXOC4_c.1141-2857:BRAFInv Protein Fusion: out of frame {EXOC4:BRAF}	NA
EVT-P-002 4206-T01-IM6-130	P-0024 206-T01-IM6		EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140487956	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) rearrangement: c.87-8072:EXOC4_c.1141-572:BRAFInv Protein Fusion: out of frame {EXOC4:BRAF}	NA
EVT-P-004 7918-T04-IM7-131	P-0047 918-T04-IM7		FAM131B-BRAF	FAM131B	in-frame	three_prime	143056532	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel Protein Fusion: in frame {FAM131B:BRAF}	NA
EVT-P-004 7918-T02-IM6-132	P-0047 918-T02-IM6		FAM131B-BRAF	FAM131B	in-frame	three_prime	143056532	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel Protein Fusion: in frame {FAM131B:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 4845-T01-I M6-134	P-0044 845-T0 1-IM6		GTF2I -BRAF	GTF2I	unknown	three_prime	140485631	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2I (NM_032999) - BRAF (NM_004333) fusion: c.768:GTF2I_c.1177+1717: BRAFInv Protein Fusion: mid-exon {GTF2I:BRAF}	NA
EVT-P-000 0828-T05-I M6-135	P-0000 828-T0 5-IM6		GTF2I RD1-BRAF	GTF2IRD1	in-frame	three_prime	140483546	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2IRD1 (NM_005685) - BRAF (NM_004333) rearrangement: c.2320+14147:GTF2IRD1_c.1178-589: BRAFInv Protein Fusion: in frame {GTF2IRD1:BRAF}	NA
EVT-P-004 6745-T01-I M6-136	P-0046 745-T0 1-IM6		KCND2-BRAF	KCND2	out-of-frame	three_prime	140485867	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KCND2 (NM_012281) - BRAF (NM_004333) fusion: c.1115+91797:KCND2_c.1177+1481:BRAFInv Protein Fusion: out of frame {KCND2:BRAF}	NA
EVT-P-005 6952-T01-I M6-138	P-0056 952-T0 1-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493821	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-141:KIAA1549_c.1140+287:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-004 4270-T02-I M6-139	P-0044 270-T0 2-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491645	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1560:KIAA1549_c.1140+2463:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 7592-T01-I M6-140	P-0027 592-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threonine_prime	140492343	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5247+769:KIAA1549_c.1140+1765:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-005 5099-T03-I M6-141	P-0055 099-T0 3-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threonine_prime	140490663	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+1357:KIAA1549_c.1141-3279:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-005 8584-T01-I M7-142	P-0058 584-T0 1-IM7		KIAA1549-B RAF	KIAA1549	in-frame	threonine_prime	140491493	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1663:KIAA1549_c.1140+2615:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-006 6030-T01-I M7-143	P-0066 030-T0 1-IM7		KIAA1549-B RAF	KIAA1549	in-frame	threonine_prime	140482785	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1172:KIAA1549_c.1314+36:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 5153-T01-I M6-144	P-0035 153-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threonine_prime	140491093	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon15 fused with BRAF exon9) : c.4930-2046:KIAA1549_c.1140+3015:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 5153-T03-I M6-145	P-0035 153-T0 3-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140491093	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2046:KIAA1549_c.1140+3015:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-004 9925-T01-I M6-146	P-0049 925-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140489497	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+301:KIAA1549_c.1141-2113:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 5844-T03-I M7-147	P-0035 844-T0 3-IM7		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140493285	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 5844-T04-I M7-148	P-0035 844-T0 4-IM7		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140493285	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-001 7693-T02-I M6-149	P-0017 693-T0 2-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-495:KIAA1549_c.1141-2153:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 2963-T01-I M7-150	P-0062 963-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482198	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-27:KIAA1549_c.1314+623:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-001 2575-T01-I M5-151	P-0012 575-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490295	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+793:KIAA1549_c.1141-2911:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-005 2411-T05-I M7-152	P-0052 411-T05-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 4117-T01-I M6-153	P-0034 117-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140483812	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4346-57:KIAA1549_c.1178-855:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-005 2411-T03-I M6-154	P-0052 411-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0009097-T01-IM5-155	P-0009097-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488817	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.5247+3837:KIAA1549_c.1141-1433:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0005121-T02-IM5-156	P-0005121-T02-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491248	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+2454:KIAA1549_c.1140+2860:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0064872-T01-IM7-157	P-0064872-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489277	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4032+4347:KIAA1549_c.1141-1893:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0013777-T01-IM5-158	P-0013777-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481674	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549(NM_001164665) - BRAF (NM_004333) rearrangement: c.5248-2734:KIAA1549_c.1315-181:BR AFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0040760-T01-IM6-159	P-0040760-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481540	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2627:KIAA1549_c.1315-47:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 4191-T01-I M6-160	P-0014 191-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140492679	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion : c.5248-320: KIAA1549_c.1140+1429:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-001 4381-T01-I M6-161	P-0014 381-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140492888	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused to BRAF exon 9 -18): c.4930-2082:KIAA1549_c.1140+1220:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-002 8611-T01-I M6-162	P-0028 611-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140482786	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-001 5540-T01-I M6-163	P-0015 540-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140489119	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.4930-69:KIAA1549_c.1141-1735:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-006 2070-T01-I M7-164	P-0062 070-T0 1-IM7		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140490443	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1064:KIAA1549_c.1141-3059:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 2404-T01-I M6-165	P-0022 404-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threepri	140482177	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 11-18) : c.5248-2270:KIAA1549_c.1314+644:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-000 4032-T01-I M5-166	P-0004 032-T0 1-IM5		KIAA1549-B RAF	KIAA1549	in-frame	threepri	140487892	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9) : c.5248-1949:KIAA1549_c.1141-508:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-004 3372-T01-I M6-167	P-0043 372-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threepri	140487523	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2204:KIAA1549_c.1141-139:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-002 8611-T02-I M6-168	P-0028 611-T0 2-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threepri	140482786	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18) : c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-004 3874-T01-I M6-169	P-0043 874-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threepri	140490001	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-2907:KIAA1549_c.1141-2617:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 9024-T04-I M7-170	P-0069 024-T0 4-IM7		KIAA1549-B RAF	KIAA1549	unknown	three_prime	140492117	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5059:KIAA1549_c.1140+1991:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-002 9361-T01-I M6-171	P-0029 361-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140489625	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-2667:KIAA1549_c.1141-2241:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-001 6557-T01-I M6-172	P-0016 557-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140488576	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955:KIAA1549_c.1141-1192:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 2982-T01-I M6-173	P-0032 982-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140491506	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10:KIAA1549_c.1140+2602:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-006 4310-T04-I M7-174	P-0064 310-T0 4-IM7		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140493446	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+3560:KIAA1549_c.1140+662:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 1127-T01-I M6-175	P-0021 127-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140491741	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-10 fused with BRAF exons 9-18) : c.4032+3996:KIAA1549_c.1140+2367:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-001 7693-T01-I M6-176	P-0017 693-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (c.1141-2153): c.4930-495: KIAA1549_c.1141-2153:BR AFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-002 0125-T01-I M6-177	P-0020 125-T0 1-IM6		KIAA1549-B RAF	KIAA1549	unknown	three_prime	140492175	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-17 fused to BRAF exons 9-18): c.5260:KIAA1549_c.1140+1933:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-004 5683-T02-I M6-178	P-0045 683-T0 2-IM6		KIAA1549-B RAF	KIAA1549	unknown	three_prime	140492850	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5189:KIAA1549_c.1140+1258:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-001 9127-T01-I M6-179	P-0019 127-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140490657	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 9-18): c.5247+2110:KIAA1549_c.1141-3273:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006 6022-T01-I M7-180	P-0066 022-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488035	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-4074:KIAA1549_c.1141-651:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 6680-T01-I M6-181	P-0036 680-T01-IM6		KLHL12-BRAF	KLHL12	in-frame	three_prime	202880970	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			KLHL12 (NM_021633) - BRAF (NM_004333) fusion (KLHL12 exons 1-4 fused to BRAF exons 9-18): t(1;7)(q32.1;q34)(chr1:g.202880970::chr7:g.140493984) Protein Fusion: in frame {KLHL12:BRAF}	NA
EVT-P-004 7833-T02-I M6-183	P-0047 833-T02-IM6		LRBA-BRAF	LRBA	in-frame	three_prime	151502269	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			LRBA (NM_001199282) - BRAF (NM_004333) fusion: t(4;7)(q31.3;q34)(chr4:g.151502269::chr7:g.140493444) Protein Fusion: in frame {LRBA:BRAF}	NA
EVT-P-000 6412-T01-I M5-184	P-0006 412-T01-IM5		LUC7L2-BRAF	LUC7L2	unknown	five_prime	140501109	6	287	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103:BRADFdel Antisense fusion	NA
EVT-P-002 6158-T04-I M6-185	P-0026 158-T04-IM6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) fusion: c.1417-9863:MAD1L1_c.1140+2359:BRAFdup Protein Fusion: in frame {MAD1L1:BRAF}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _nam e	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mai ns	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-002 6158-T01-I M6-186	P-0026 158-T0 1-IM6		MAD1 L1-BR AF	MA D1L 1	in-fr ame	thr ee_ pr im e	140491 749	8	380	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		MAD1L1 (NM_001013837) - BRAF (NM_004333) Rearrangement : c.1417-98 63_c.1140+2359dup Protein Fusion: in frame {MAD1L1:BRAF}	NA
EVT-P-000 6601-T03-I M6-187	P-0006 601-T0 3-IM6		MIND Y4-BR AF	MIN DY4	in-fr ame	thr ee_ pr im e	140490 696	9	381	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		FAM188B (NM_032222) - BRAF(NM_004333) Fusion (FAM188B exon5 fused with BRAF exon9) : c.664-366:F AM188B_c.1141-3312:BR AF Protein Fusion: in frame {FAM188B:BRAF}	NA
EVT-P-002 3220-T01-I M6-188	P-0023 220-T0 1-IM6		MKLN 1-BRA F	MKL N1	in-fr ame	thr ee_ pr im e	140496 796	8	327	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		MKLN1 (NM_013255) - BRAF (NM_004333) rearrangement: c.98+4070: MKLN1_c.981-2529:BRAFin v Protein Fusion: in frame {MKLN1:BRAF}	NA
EVT-P-003 5393-T02-I M6-189	P-0035 393-T0 2-IM6		MKRN 1-BRA F	MK RN1	in-fr ame	thr ee_ pr im e	140488 436	9	381	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.771+833:MKRN1_c.1141- 1052:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-000 9024-T01-I M5-190	P-0009 024-T0 1-IM5		MKRN 1-BRA F	MK RN1	in-fr ame	thr ee_ pr im e	140493 485	8	380	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G: MKRN1_c.1140+623:BRAF dup Protein fusion: in frame (MKRN1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008083-T03-IM6-191	P-0008083-T03-IM6		MKRN1-BRAF	MKRN1	out-of-frame	three_prime	140484511	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.315-432:MKRN1_c.1178-1554:BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA
EVT-P-0001036-T01-IM3-192	P-0001036-T01-IM3		MKRN1-BRAF	MKRN1	in-frame	three_prime	140482250	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0047135-T01-IM6-193	P-0047135-T01-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140492510	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-972:MKRN1_c.1140+1598:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-0004373-T02-IM5-194	P-0004373-T02-IM5		MKRN1-BRAF	MKRN1	unknown	three_prime	140434466	18	744	False	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.314+4351:MKRN1_c.2232:BRAFdup Protein fusion: mid-exon (MKRN1-BRAF)	NA

event_id	sampl_e_id	pa_tie_nt_id	fusion_nam_e	part_ner_ge_ne	fra_me_st_atu_s	tar_get_ro_le	breakp_o_int_ge_nomic	break_p_o_int_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 9024-T02-I M6-195	P-0009 024-T0 2-IM6		MKRN 1-BRA F	MK RN1	in-fr ame	thr_ee _pr im_e	140493 485	8	380	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-185:MKRN1_c.1140+ 623:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-003 1896-T02-I M6-196	P-0031 896-T0 2-IM6		MKRN 1-BRA F	MK RN1	in-fr ame	thr_ee _pr im_e	140482 581	10	438	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-4 fused to BRAF exons 11-18): c.772- 1018:MKRN1_c.1314+240: BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-000 8083-T02-I M6-197	P-0008 083-T0 2-IM6		MKRN 1-BRA F	MK RN1	out-of-fr ame	thr_ee _pr im_e	140484 505	10	393	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-2 fused with BRAF exons 10-18): c. 315-428:MKRN1_c.1178-15 48:BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA
EVT-P-001 3536-T01-I M5-198	P-0013 536-T0 1-IM5		MKRN 1-BRA F	MK RN1	in-fr ame	thr_ee _pr im_e	140483 618	10	393	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.544+172: MKRN1_c.1178-661:BRAFd up Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-004 8036-T01-I M6-199	P-0048 036-T0 1-IM6		NEO1 -BRA F	NE O1	in-fr ame	thr_ee _pr im_e	140497 651	7	327	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		NEO1 (NM_002499) - BRAF (NM_004333) fusion: t(7;15)(q34;q24.1)(chr7:g.14 0497651::chr15:g.73575682) Protein Fusion: in frame {NEO1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 3972-T03-I M6-200	P-0043 972-T0 3-IM6		NRF1- BRAF	NRF 1	in-frame	three_prime	140487 712	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.607-7410:NRF1_c.1141-328:BR AFInv Protein Fusion: in frame {NRF1:BRAF}	NA
EVT-P-006 5446-T01-I M7-201	P-0065 446-T0 1-IM7		NRF1- BRAF	NRF 1	in-frame	three_prime	140483 810	10	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.1349-10893:NRF1_c.1178-853:BR AFInv Protein Fusion: in frame {NRF1:BRAF}	NA
EVT-P-005 9002-T01-I M7-202	P-0059 002-T0 1-IM7		OSBP L7-BR AF	OSB PL7	in-frame	three_prime	140491 530	8	380	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		OSBPL7 (NM_145798) - BRAF (NM_004333) fusion: t(7;17)(q34;q21.32)(chr7:g.140491530::chr17:g.45890381) Protein Fusion: in frame {OSBPL7:BRAF}	NA
EVT-P-005 4958-T02-I M6-203	P-0054 958-T0 2-IM6		PAK2- BRAF	PAK 2	in-frame	three_prime	196530 579	1	1	True	retained	1.0	False	Protein kinase domain; RAS- binding domain; Cy- steine- rich do- main			PAK2 (NM_002577) - BRAF (NM_004333) rearrangement: t(3;7)(q29;q34)(chr3:g.196530579::chr7:g.140486065) Protein Fusion: in frame {PAK2:BRAF}	NA
EVT-P-000 8789-T01-I M5-205	P-0008 789-T0 1-IM5		PHTF 2-BR AF	PHT F2	in-frame	three_prime	140489 236	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		PHTF2 (NM_001127358) - BRAF (NM_004333) rearrangement: c.2224-367:PHTF2_c.1141-1852:BR AFInv Protein fusion: in frame (PHTF2-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 4819-T01-I M6-206	P-0044 819-T0 1-IM6		PLPP 1-BRAF	PLP P1	out-of-frame	threep prime	547891 92	18	767	True	lost	0.0	False		Protein kinase domain; RAS- binding domain; Cysteine- rich domain		PPAP2A (NM_176895) - BRAF (NM_004333) fusion: t(5;7)(q11.2;q34)(chr5:g.547 89192::chr7:g.140482123) Protein Fusion: out of frame {PPAP2A:BRAF}	NA
EVT-P-000 9694-T01-I M5-207	P-0009 694-T0 1-IM5		PRKAR 1B-BRAF	PRKAR 1B	in-frame	threep prime	140491 855	8	380	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		PRKAR1B (NM_001164758) -BRAF (NM_004333) Rearrangement : c.892-255 1:PRKAR1B_c.1140+2253: BRAFDup Protein fusion: in frame (PRKAR1B-BRAF)	NA
EVT-P-000 8567-T01-I M5-208	P-0008 567-T0 1-IM5		PRKAR 2B-BRAF	PRKAR 2B	in-frame	threep prime	140486 416	9	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063 :PRKAR2B_c.1177+932:BR AFInv Protein fusion: in frame (PRKAR2B-BRAF)	NA
EVT-P-004 1181-T01-I M6-209	P-0041 181-T0 1-IM6		PTPRZ 1-BRAF	PTPRZ 1	in-frame	threep prime	140492 428	8	380	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		PTPRZ1 (NM_002851) - BRAF (NM_004333) fusion: c.5367+457:PTPRZ1_c.114 0+1680:BRAFInv Protein Fusion: in frame {PTPRZ1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 2884-T01-IM5-210	P-0012 884-T01-IM5		PWWP2A-BRAF	PWWP2A	in-frame	three_prime	159539324	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			PWWP2A (NM_001130864) - BRAF(NM_004333) Rearrangement : t(5;7)(q33.3;q34)(chr5:g.159539324::chr7:g.140495243) Protein fusion: in frame (PWWP2A-BRAF)	NA
EVT-P-006 8260-T01-IM7-211	P-0068 260-T01-IM7		RBMS3-BRAF	RBMS3	in-frame	three_prime	30030473	18	767	True	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		RBMS3 (NM_001003793) - BRAF (NM_004333) fusion: t(3;7)(p24.1;q34)(chr3:g.30030473::chr7:g.140482707) Protein Fusion: in frame {RBMS3:BRAF}	NA
EVT-P-006 4764-T01-IM7-212	P-0064 764-T01-IM7		RGS3-BRAF	RGS3	out-of-frame	three_prime	140493282	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RGS3 (NM_144488) - BRAF (NM_004333) fusion: t(7;9)(q34;q32)(chr7:g.140493282::chr9:g.116291333) Protein Fusion: out of frame {RGS3:BRAF}	NA
EVT-P-004 1308-T02-IM6-213	P-0041 308-T02-IM6		RRBP1-BRAF	RRBP1	in-frame	three_prime	140485917	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RRBP1 (NM_004587) - BRAF (NM_004333) fusion: t(7;20)(q34;p12.1)(chr7:g.140485917::chr20:g.17634546) Protein Fusion: in frame {RRBP1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007392-T01-IM5-214	P-0007392-T01-IM5		SCRIB-BRAF	SCRIB	in-frame	three_prime	140482150	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SCRIB (NM_182706) - BRAF (NM_004333) rearrangement: t(7;8)(q34;q24.3)(chr7:g.140482150::chr8:g.144876675) Protein fusion: in frame (SCRIB-BRAF)	NA
EVT-P-0056639-T01-IM6-215	P-0056639-T01-IM6		SETBP1-BRAF	SETBP1	in-frame	three_prime	140486594	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SETBP1 (NM_015559) - BRAF (NM_004333) fusion: t(7;18)(q34;q12.3)(chr7:g.140486594::chr18:g.42638299) Protein Fusion: in frame {SETBP1:BRAF}	NA
EVT-P-0048950-T01-IM6-216	P-0048950-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140492033	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+30037:SND1_c.1140+2075:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0012395-T01-IM5-217	P-0012395-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140493527	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1152+27174:SND1_c.1140+581:BRAFinv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0009619-T01-IM5-218	P-0009619-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140482494	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1968+2394:SND1_c.1314+327:BRAFinv Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 2299-T04-I M6-219	P-0052 299-T0 4-IM6		SND1- BRAF	SND 1	unknown	three_prime	140482 935	10	400	False	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1038+2929:SND1_c.1200:BRAFInv Protein Fusion: mid-exon {SND1:BRAF}	NA
EVT-P-000 1687-T01-I M3-220	P-0001 687-T0 1-IM3		SND1- BRAF	SND 1	in-frame	three_prime	140490 224	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) inversion: c.1152+28381_c.1141-2840inv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-000 4103-T01-I M5-221	P-0004 103-T0 1-IM5		SND1- BRAF	SND 1	in-frame	three_prime	140481 685	11	439	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1038+5799:SND1_c.1315-192:BRAFInv. Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-002 9727-T01-I M6-222	P-0029 727-T0 1-IM6		SND1- BRAF	SND 1	in-frame	three_prime	140482 257	10	438	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+12095:SND1_c.1314+564:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-006 8194-T01-I M7-223	P-0068 194-T0 1-IM7		SND1- BRAF	SND 1	in-frame	three_prime	140489 175	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+17686:SND1_c.1141-1791:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 8140-T01-I M5-224	P-0008 140-T0 1-IM5		SND1- BRAF	SND 1	in-frame	three_prime	140489 439	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18) : c.1039-3849:SNV Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-003 4168-T01-I M6-225	P-0034 168-T0 1-IM6		SND1- BRAF	SND 1	in-frame	three_prime	140489 421	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused to BRAF exons 9-18) : c.1152+6972:SNV Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-002 5075-T01-I M6-226	P-0025 075-T0 1-IM6		SND1- BRAF	SND 1	in-frame	three_prime	140490 546	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1- 9 with BRAF exons 9-18) : c.1039-2471:SNV Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-000 2071-T02-I M5-227	P-0002 071-T0 2-IM5		SND1- BRAF	SND 1	in-frame	three_prime	140489 292	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152+18606:SNV Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-000 2197-T01-I M3-228	P-0002 197-T0 1-IM3		SND1- BRAF	SND 1	in-frame	five_prime	140488 303	9	381	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		c.1528-8436:SNV Protein fusion: in frame (BRAF-SND1)	NA

event_id	sampl_e_id	pa_tie_nt_i_d	fusion_nam_e	part_ner_ge_ne	fra_me_st_atu_s	tar_get_ro_le	breakp_o_int_ge_nomic	break_p_o_int_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 7533-T01-I M5-229	P-0007 533-T0 1-IM5		SND1- BRAF	SND 1	in-fr ame	thr_ee _pr im_e	140482 226	10	438	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738:SND1_c.1314+595:BRAFinv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-003 6567-T01-I M6-230	P-0036 567-T0 1-IM6		SND1- BRAF	SND 1	in-fr ame	thr_ee _pr im_e	140493 552	8	380	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		SND1 (NM_014390) - BRAF (NM_004333) Fusion (SND1 exon10 fused with BRAF exon9) : c.1153-22013:SND1_c.1140+556:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-003 4845-T01-I M6-231	P-0034 845-T0 1-IM6		SND1- BRAF	SND 1	in-fr ame	thr_ee _pr im_e	140489 241	9	381	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1527+3658:SND1_c.1141-1857:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-005 4364-T01-I M6-232	P-0054 364-T0 1-IM6		SNX8- BRAF	SNX 8	in-fr ame	thr_ee _pr im_e	140488 730	9	381	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		SNX8 (NM_013321) - BRAF (NM_004333) fusion: c.1284+565:SNX8_c.1141-1346:BRAFdup Protein Fusion: in frame {SNX8:BRAF}	NA
EVT-P-002 6773-T01-I M6-234	P-0026 773-T0 1-IM6		SVOP L-BRA F	SVO PL	unk now n	five _pr im_e	140482 097	11	439	True	lost	0.0	False	RAS-bi nding domai n; Cyst eine-ri ch do main	Prot ein kina se do ma in		BRAF (NM_004333) rearrangement: c.1315-604: BRAF_chr7:g.138277426del Transcript Fusion {BRAF:SVOPL}	NA
EVT-P-005 1214-T01-I M6-235	P-0051 214-T0 1-IM6		TMPR SS2-B RAF	TMP RSS 2	in-fr ame	thr_ee _pr im_e	140502 592	6	238	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain	Cystein e-rich domain	TMPrSS2 (NM_001135099) - BRAF (NM_004333) fusion: t(7;21)(q34;q22.3)(chr7:g.140502592::chr21:g.42867846) Protein Fusion: in frame {TMPrSS2:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 8524-T01-I M6-236	P-0018 524-T01-IM6		TPRSS2-BRAF	TPRSS2	out-of-frame	threep_rime	140490337	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(21;7)(q22.3;q34)(chr21:g.42873801::chr7:g.140490337) Protein Fusion: out of frame {TPRSS2:BRAF}	NA
EVT-P-002 2778-T01-I M6-237	P-0022 778-T01-IM6		TPRSS2-BRAF	TPRSS2	in-frame	threep_rime	140483147	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(7;21)(q34;q22.3)(chr7:g.140483147::chr21:g.42878796) Protein Fusion: in frame {TPRSS2:BRAF}	NA
EVT-P-004 3836-T01-I M6-239	P-0043 836-T01-IM6		TRA2B-BRAF	TRA2B	in-frame	threep_rime	185650761	1	1	True	retained	1.0	False	Protein kinase domain; RAS-binding domain; Cysteine-rich domain			TRA2B (NM_004593) - BRAF (NM_004333) fusion: t(3;7)(q27.2;q34)(chr3:g.185650761::chr7:g.140491978) Protein Fusion: in frame {TRA2B:BRAF}	NA
EVT-P-005 4430-T02-I M6-240	P-0054 430-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	threep_rime	140487875	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2719-489;TRIM24_c.1141-491;BRAFinv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-002 5699-T02-I M6-241	P-0025 699-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	threep_rime	140486854	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2014+674;TRIM24_c.1177+494;BRAFinv Protein Fusion: in frame {TRIM24:BRAF}	NA

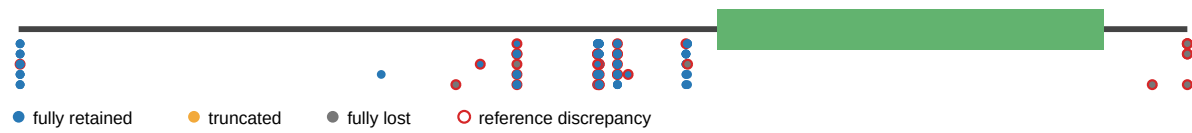
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EVT-P-004 5111-T01-I M6-242	P-0045 111-T0 1-IM6		TRIM2 4-BRA F	TRI M24	in-fr ame	thr_ee _pr im e	140488 049	9	381	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2257-760:TRIM24_c.1141 -665:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-006 6835-T01-I M7-243	P-0066 835-T0 1-IM7		TRIM2 4-BRA F	TRI M24	in-fr ame	thr_ee _pr im e	140490 589	9	381	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1879-502:TRIM24_c.1141 -3205:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-006 8451-T01-I M7-244	P-0068 451-T0 1-IM7		TRIM2 4-BRA F	TRI M24	in-fr ame	thr_ee _pr im e	140482 765	10	438	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.484-860:TRIM24_c.1314+ 56:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-003 0988-T01-I M6-245	P-0030 988-T0 1-IM6		TRIM2 4-BRA F	TRI M24	unk now n	thr_ee _pr im e	140491 002	8	380	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2349:TRIM24_c.1140+31 06:BRAFInv Protein Fusion: mid-exon {TRIM24:BRAF}	NA
EVT-P-003 2569-T01-I M6-246	P-0032 569-T0 1-IM6		TRIM2 4-BRA F	TRI M24	in-fr ame	thr_ee _pr im e	140487 936	9	381	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		TRIM24 (NM_015905) - BRAF (NM_004333) Rearrangement : c.1878+79 1:TRIM24_c.1141-552inv Protein Fusion: in frame {TRIM24:BRAF}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _nam e	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_ retained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mai ns	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-003 2569-T02-I M6-247	P-0032 569-T0 2-IM6		TRIM2 4-BRA F	TRI M24	in-fr ame	thr ee_ pr im e	140487 936	9	381	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1878+791:TRIM24_c.114 1-552:BRAFinv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-001 9039-T01-I M6-248	P-0019 039-T0 1-IM6		UBE2 H-BR AF	UBE 2H	out- of-fr ame	thr ee_ pr im e	140499 248	7	327	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		BRAF (NM_004333) - UBE2H (NM_003344) rearrangement: c.980+914: BRAF_c.299-4611:UBE2Hd up Protein Fusion: out of frame {UBE2H:BRAF}	NA
EVT-P-002 7762-T01-I M6-250	P-0027 762-T0 1-IM6		ZC3H AV1-B RAF	ZC3 HAV 1	in-fr ame	thr ee_ pr im e	140485 946	9	393	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		ZC3HAV1 (NM_020119) - BRAF (NM_004333) rearrangement: c.698-431:Z C3HAV1_c.1177+1402:BRA Fdup Protein Fusion: in frame {ZC3HAV1:BRAF}	NA
EVT-P-000 8920-T01-I M5-251	P-0008 920-T0 1-IM5		ZNF2 07-BR AF	ZNF 207	in-fr ame	thr ee_ pr im e	140485 847	9	393	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34; q11.2)(chr7:g.140485847::c hr17:g.30687533) Protein fusion: in frame (ZNF207-BRAF)	NA

Figures

domain retention outliers

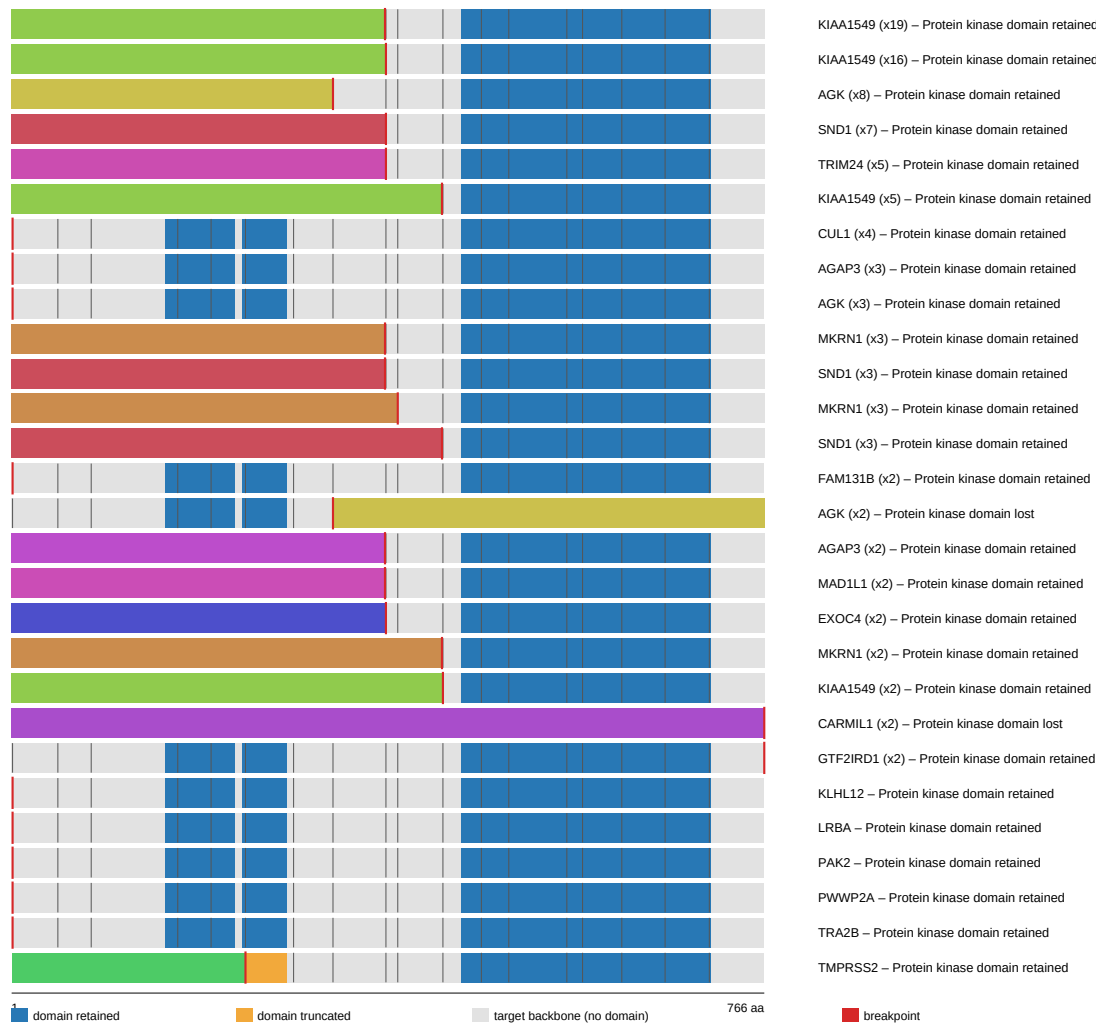
BRAF domain-retention track



fusion schematic

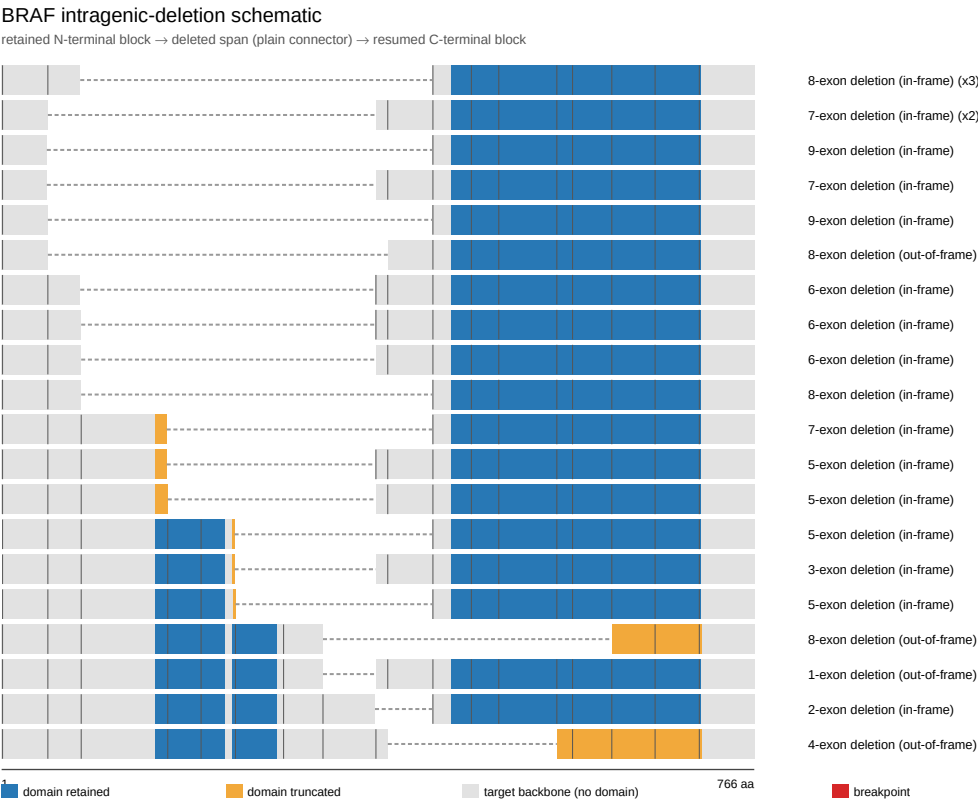
BRAF fusion-transcript schematic

partner block → breakpoint → retained BRAF portion (domain-colored, exon ticks)



Showing the top 28 of 100 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

